# PF3 Worked Examples: Inverse Trigonometric Functions

Recording companion for the PF3 CheckIt support video — Math 172, Santa Ana College

## How to use this document

- Each worked example is on its own page, in the same order as the Try It tabs in the interactive explainer, with the same numbers.
- The right-hand third of every page is intentionally blank. It is space for handwriting or ink notation while you narrate (for example, after inserting this PDF as a printout in OneNote).
- Say it at the start of the video: your CheckIt problem will have different numbers than the examples here. The reasoning is what transfers.

## The one question behind every example

**Which angle has this sine, cosine, or tangent, and is that angle inside the function's range?** Find an angle on the unit circle with the right value. Then check the range: arcsine and arctangent give angles from negative pi over 2 to pi over 2, the right half of the circle. Arccosine gives angles from 0 to pi, the top half. If your angle is outside, find the one inside that has the same value.

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# Example 1: Evaluate 1

$$\arcsin\left(\frac{1}{2}\sqrt{3}\right)$$

## Question

What is the exact value?

**A.** $\pi/6$
**B.** $(2\pi)/3$
**C.** $\pi/3$
**D.** $-\pi/3$

## Answer and reasoning

The input is $(\sqrt{3})/2$. Which angle has sine $(\sqrt{3})/2$? Both $\pi/3$ and $(2\pi)/3$ do, but arcsine only gives angles from $-\pi/2$ to $\pi/2$, so $(2\pi)/3$ is out. The angle $\pi/6$ has cosine $(\sqrt{3})/2$, not sine. **The value is $\boldsymbol{\pi/3}$.**

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# Example 2: Evaluate 2

$$\arccos\left(-\frac{1}{2}\sqrt{2}\right)$$

## Question

What is the exact value?

- **A.** $-\pi/4$
- **B.** $(3\pi)/4$
- **C.** $(5\pi)/4$
- **D.** $\pi/4$

## Answer and reasoning

The input is $-(\sqrt{2})/2$. Arccosine gives angles from 0 to $\pi$, the top half of the circle. Cosine is negative on the left side, so the angle is in quadrant II: $(3\pi)/4$. The angle $(5\pi)/4$ also has that cosine but is on the bottom half. Arccosine is never negative, which rules out $-\pi/4$. **The value is $(3\pi)/4$.**

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# Example 3: Evaluate 3

$$\arctan(-\sqrt{3})$$

## Question

What is the exact value?

**A.** $(2\pi)/3$
**B.** $-\pi/6$
**C.** $(5\pi)/3$
**D.** $-\pi/3$

## Answer and reasoning

Tangent is $\sqrt{3}$ at $\pi/3$. Arctangent gives angles between $-\pi/2$ and $\pi/2$, and a negative input gives a negative angle, so the answer is $-\pi/3$. The angles $(2\pi)/3$ and $(5\pi)/3$ have the same tangent but are outside the range. Tangent at $-\pi/6$ is $-(\sqrt{3})/3$. **The value is $-\pi/3$.**

# Example 4: Evaluate 4

$$\arcsin(-1/2)$$

## Question

What is the exact value?

- **A.** $-\pi/6$
- **B.** $(7\pi)/6$
- **C.** $(11\pi)/6$
- **D.** $-\pi/3$

## Answer and reasoning

Sine is $1/2$ at $\pi/6$. A negative input to arcsine gives a negative angle, in quadrant IV: $-\pi/6$. The angle $(11\pi)/6$ points the same way, but arcsine's range only goes down to $-\pi/2$, so the answer is written as $-\pi/6$. Sine at $-\pi/3$ is $-(\sqrt{3})/2$. **The value is $-\pi/6$.**

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# Example 5: Value 1

$$\arccos(-1/2)$$

## Question

What is the exact value?

**A.** $-\pi/3$
**B.** $(4\pi)/3$
**C.** $(2\pi)/3$
**D.** $\pi/3$

## Answer and reasoning

Cosine is $1/2$ at $\pi/3$. A negative cosine means the left side of the circle, and arccosine stays on the top half, so the angle is in quadrant II: $\pi-\pi/3=(2\pi)/3$. The answer $-\pi/3$ uses arcsine's rule for negatives, which does not work for arccosine. **The value is $(2\pi)/3$.**

# Example 6: Value 2

$$\arctan\left(\frac{1}{3}\sqrt{3}\right)$$

## Question

What is the exact value?

**A.** $\pi/3$
**B.** $\pi/6$
**C.** $(7\pi)/6$
**D.** $\pi/4$

## Answer and reasoning

The input is $(\sqrt{3})/3$, which is the same as $1/(\sqrt{3})$. At $\pi/6$, sine is $1/2$ and cosine is $(\sqrt{3})/2$, so tangent is $1/(\sqrt{3})$. At $\pi/3$ tangent is $\sqrt{3}$ instead, and at $\pi/4$ it is 1. The angle $(7\pi)/6$ has the same tangent but is outside the range. **The value is $\pi/6$.**

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# Example 7: Value 3

$$\arccos(0)$$

## Question

What is the exact value?

**A.** $0$
**B.** $(3\pi)/2$
**C.** $-\pi/2$
**D.** $\pi/2$

## Answer and reasoning

Cosine is the x-coordinate on the unit circle, and it is 0 at the top and bottom of the circle. Arccosine only uses the top half, from 0 to $\pi$, so the answer is the top: $\pi/2$. The answer 0 is arcsine of 0, not arccosine. **The value is $\pi/2$.**

# Example 8: Value 4

$$\arcsin(-1/2\sqrt{2})$$

## Question

What is the exact value?

**A.** $(5\pi)/4$
**B.** $(7\pi)/4$
**C.** $-\pi/4$
**D.** $(3\pi)/4$

## Answer and reasoning

The input is $-(\sqrt{2})/2$. Sine is $(\sqrt{2})/2$ at $\pi/4$, and a negative input to arcsine gives a negative angle: $-\pi/4$. The angles $(5\pi)/4$ and $(7\pi)/4$ also have that sine, but they are outside arcsine's range. **The value is $-\pi/4$.**

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# Example 9: Range 1

$$f(x)=\arcsin(x)$$

## Question

What is the range of this function?

**A.** $[0,\pi]$
**B.** $[-\pi/2,\pi/2]$
**C.** $(-\pi/2,\pi/2)$
**D.** $[-1,1]$

## Answer and reasoning

Arcsine gives angles on the right half of the unit circle, from $-\pi/2$ to $\pi/2$, and it reaches both ends: $\arcsin(1)=\pi/2$ and $\arcsin(-1)=-\pi/2$. So the brackets are square. The interval from $-1$ to $1$ is the domain, the inputs. **The range is $[-\pi/2,\pi/2]$.**

# Example 10: Domain 1

$$g(x)=\arccos(x)$$

## Question

What is the domain of this function?

**A.** $[-1,1]$
**B.** $[0,\pi]$
**C.** $(-\infty,\infty)$
**D.** $[-\pi/2,\pi/2]$

## Answer and reasoning

The input to arccosine is a cosine value, and cosine is always between $-1$ and $1$. So only those inputs make sense: $\arccos(2)$ has no answer. The interval from 0 to $\pi$ is the range, the outputs. **The domain is $[-1,1]$.**

# Example 11: Range 2

$$h(x)=\arctan(x)$$

## Question

What is the range of this function?

**A.** $[-\pi/2,\pi/2]$
**B.** $(-\infty,\infty)$
**C.** $(-\pi/2,\pi/2)$
**D.** $[0,\pi]$

## Answer and reasoning

Arctangent gives angles on the right half of the circle, like arcsine. But tangent is undefined at $\pm\pi/2$, so arctangent never reaches those angles; its graph only gets closer and closer to them. The ends are left out, so the brackets are round. All real numbers is the domain. **The range is $(-\pi/2,\pi/2)$.**

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# Example 12: Domain 2

$$h(x)=\arctan(x)$$

## Question

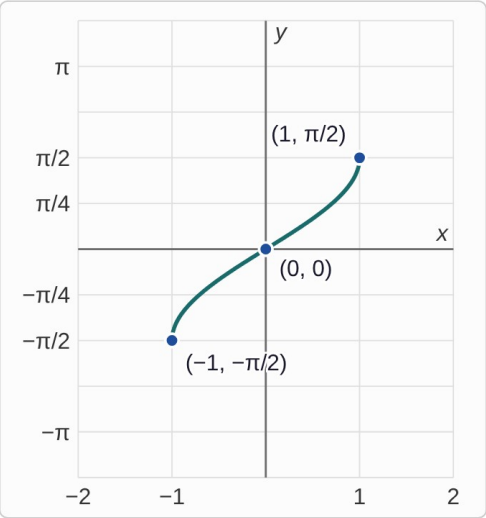
What is the domain of this function?

- **A.** $[-1,1]$
- **B.** $(-\pi/2,\pi/2)$
- **C.** $[0,\infty)$
- **D.** $(-\infty,\infty)$

## Answer and reasoning

Tangent can be any real number, so arctangent accepts any input: $\arctan(1000)$ is just an angle a little less than $\pi/2$. Only arcsine and arccosine are limited to inputs from $-1$ to $1$. The open interval from $-\pi/2$ to $\pi/2$ is the range. **The domain is all real numbers, $(-\infty,\infty)$.**

# Example 13: Graph 1




**Plotted data:** a curve that exists only from x = −1 to x = 1, with solid dots at both ends: (−1, −π/2) and (1, π/2). It passes through (0, 0) and rises from left to right. It is steepest near its two ends and flattest in the middle. Nothing is drawn to the left of x = −1 or to the right of x = 1.

## Question

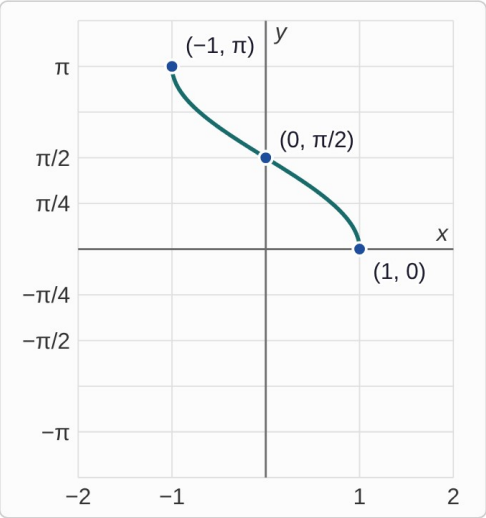
Which function has this graph?

- **A.** $y=\arccos(x)$
- **B.** $y=\arctan(x)$
- **C.** $y=\sin(x)$
- **D.** $y=\arcsin(x)$

## Answer and reasoning

The curve only exists for inputs from $-1$ to $1$, so it is arcsine or arccosine. It rises and passes through $(0, 0)$: $\arcsin(0)=0$, while $\arccos(0)=\pi/2$. The outputs run from $-\pi/2$ to $\pi/2$, the range of arcsine. Sine would repeat forever left and right. **The graph is $y=\arcsin(x)$.**

# Example 14: Graph 2




**Plotted data:** a curve that exists only from x = −1 to x = 1, with solid dots at both ends: (−1, π) and (1, 0). It passes through (0, π/2) and falls from left to right. It is steepest near its two ends. Nothing is drawn to the left of x = −1 or to the right of x = 1.

## Question

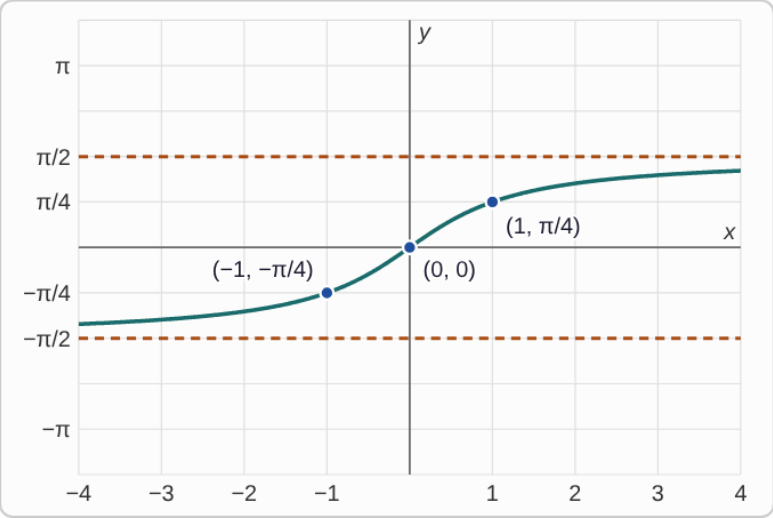
Which function has this graph?

- **A.** *y=arccos(x)*
- **B.** *y=arcsin(x)*
- **C.** *y=arctan(x)*
- **D.** *y*=sin*(x)*

## Answer and reasoning

The inputs run from *−1* to *1*, and every output is between 0 and *π*, never negative. That is the range of arccosine. Check the points: *arccos(1)=0*, *arccos(0)=π/2*, and *arccos(−1)=π*. Arcsine would rise instead. **The graph is *y=arccos(x)*.**

# Example 15: Graph 3




**Plotted data:** dashed horizontal lines at y = −π/2 and y = π/2. A curve runs across the whole grid, from x = −4 to x = 4, rising from left to right. It passes through (−1, −π/4), (0, 0), and (1, π/4). Far to the left it comes very close to the lower dashed line, and far to the right very close to the upper one, without touching either.

## Question

Which function has this graph?

**A.** $y=\arcsin(x)$
**B.** $y=\sin(x)$
**C.** $y=\arctan(x)$
**D.** $y=\arccos(x)$

## Answer and reasoning

The curve accepts every input, so it cannot be arcsine or arccosine, which stop at $-1$ and $1$. It levels off toward $y=\pm\pi/2$ without reaching them, which is the range of arctangent. Check: $\arctan(1)=\pi/4$. Sine would wave up and down between $-1$ and $1$.
**The graph is $y=\arctan(x)$.**

# Example 16: Slip 1

$$\arcsin((\sqrt{3})/2)=(2\pi)/3$$

## Question

A student wrote this because the sine of 2π/3 is the square root of 3 over 2. What is the mistake, if any?

**A.** Arcsine stops at π/2, so the answer is π/3.
**B.** The sine of 2π/3 is really one half, so the angle is wrong.
**C.** The answer should be 4π/3, the other angle with this sine.
**D.** There is no mistake.

## Answer and reasoning

It is true that $\sin((2\pi)/3)=(\sqrt{3})/2$, but $(2\pi)/3$ is outside arcsine's range. Arcsine picks the one angle on the right half of the circle with that sine. The sine of $(4\pi)/3$ is negative. **The correct value is $\pi/3$.**

# Example 17: Slip 2

$$\arccos(-(\sqrt{2})/2) = -\pi/4$$

## Question

A student wrote this. What is the mistake, if any?

**A.** There is no mistake.
**B.** The answer should be 5π/4, since cosine is negative there.
**C.** Arccosine is never negative; the answer is 3π/4.
**D.** The answer should be π/4, because arccosine ignores signs.

## Answer and reasoning

Arccosine gives angles from 0 to $\pi$, the top half of the circle, so it is never negative. A negative cosine puts the angle on the left side of the top half: $(3\pi)/4$. The student used arcsine's rule. The angle $(5\pi)/4$ is on the bottom half. **The correct value is $(3\pi)/4$.**

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# Example 18: Slip 3

$$\arcsin(2)$$

## Question

A student says this has no value. What is the mistake, if any?

**A.** It equals π/2, because 2 is close to 1.
**B.** There is no mistake.
**C.** It equals one divided by the sine of 2.
**D.** It equals 2π, one full turn around the circle.

## Answer and reasoning

Sine is never more than 1, so no angle has a sine of 2. The domain of arcsine is $[-1,1]$, and 2 is outside it. The student is right. Arcsine is not one divided by sine. **There is no mistake: arcsine of 2 is undefined.**

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# Example 19: Slip 4

$$h(x)=\arctan(x)$$

## Question

A student says the range of this function is the closed interval from negative π/2 to π/2, with square brackets. What is the mistake, if any?

**A.** The range should be all real numbers.
**B.** The range should be from 0 to π, the same as arccosine.
**C.** There is no mistake.
**D.** The ends are never reached, so use round brackets.

## Answer and reasoning

Tangent is undefined at $\pm\pi/2$, so arctangent never outputs those angles. Its graph only levels off toward them. Leaving the ends out takes round brackets. All real numbers is the domain, not the range. **The range is $(-\pi/2,\pi/2)$.**

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# Example 20: Slip 5

$$\sin^{-1}(x) = 1/(\sin(x))$$

## Question

A student wrote this. What is the mistake, if any?

**A.** There is no mistake.
**B.** The right side should be 1 over the cosine of x.
**C.** The negative 1 means arcsine here, not one over sine.
**D.** The right side should be the negative of the sine of x.

## Answer and reasoning

On a trig function, the $-1$ is not an exponent: it names the inverse function, arcsine, which takes a value and gives back an angle. One over sine is cosecant, a different function. For example, arcsine of 1 is $\pi/2$, but one over sine of 1 is about 1.19. **The notation means arcsine, not one over sine.**

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